Topic 2 Review Test

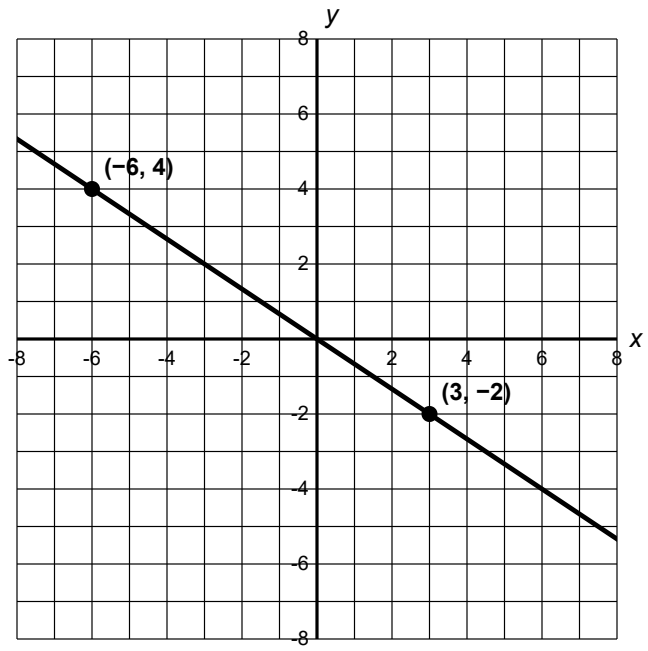
Algebra I · Module 2, Topic 2: Transforming and Comparing Linear Functions · Monday, Nov 2

Name _____ Period _____ Date _____

Directions: This review has the same kinds of questions as the Topic 2 test. Read each question carefully. Show your work. **Select TWO** means choose exactly two answers. Finish Part 1 first; Part 2 is extra practice.

Part 1 · Just like the test (Questions 1–10)

1. The graph of linear function *m* is shown on the grid.



Given $f(x) = x$ and $m(x) = af(x)$, what is the value of a?

a = _____

2. The cost of renting a bounce house, y, for x hours at a birthday party can be modeled using a linear function. The rental company charges a fixed fee of $75, plus an additional $20 per hour. The bounce house can be rented for a maximum of 5 hours per day. For one day of rental, what is the range of the function for this situation?

A. $0 \leq x \leq 5$

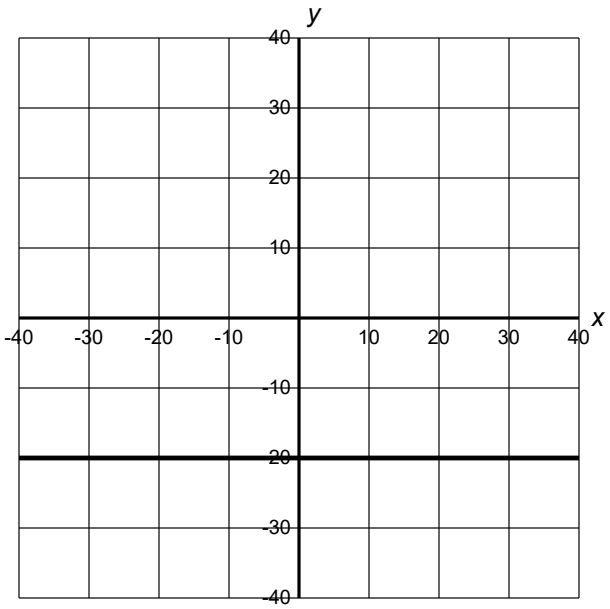
B. $20 \leq y \leq 175$

C. $75 \leq y \leq 175$

D. $0 \leq x \leq 7$

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3. What are the equation and slope of the line shown on the grid?



F. $x = -20$; slope is zero

G. $y = -20$; slope is undefined

H. $y = -20$; slope is zero

J. $x = -20$; slope is undefined

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4. Marcus graphed $f(x)$ and $g(x) = f(x - 3)$ on the same coordinate grid. Which statement describes how the graphs of f and g are related?

A. The graph of the function $f(x)$ is translated up 3 units to produce $g(x)$.

B. The graph of the function $f(x)$ is translated down 3 units to produce $g(x)$.

C. The graph of the function $f(x)$ is translated right 3 units to produce $g(x)$.

D. The graph of the function $f(x)$ is translated left 3 units to produce $g(x)$.

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5. What is the y-intercept of the linear function created by translating $f(x)$, represented in the table, 2 units right?

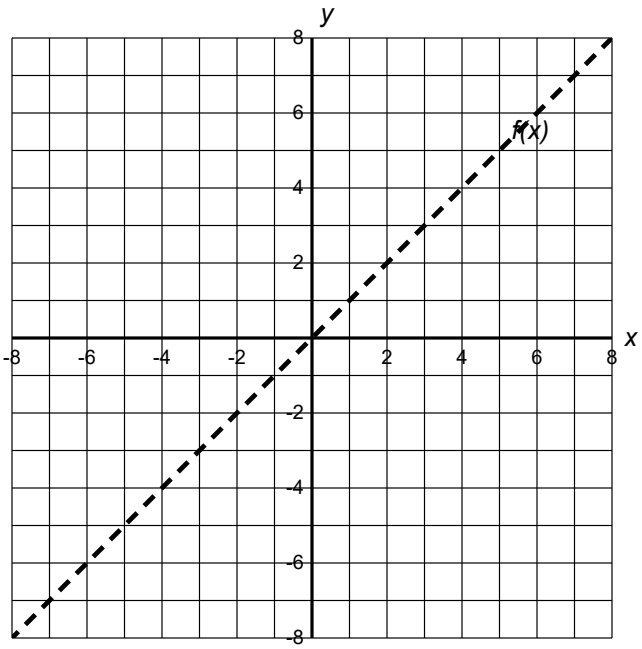
x	$f(x)$
−2	−7
−1	−4
0	−1
1	2
2	5
3	8

y-intercept = _____

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6. Graph $g(x)$ based on the given transformation of $f(x)$.

$f(x) = x$ $g(x) = f(2x)$



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7. What is the equation in slope-intercept form of the line that crosses the x-axis at −6 and is perpendicular to the line represented by $y = \frac{3}{2}x - 1$?

Equation: _____

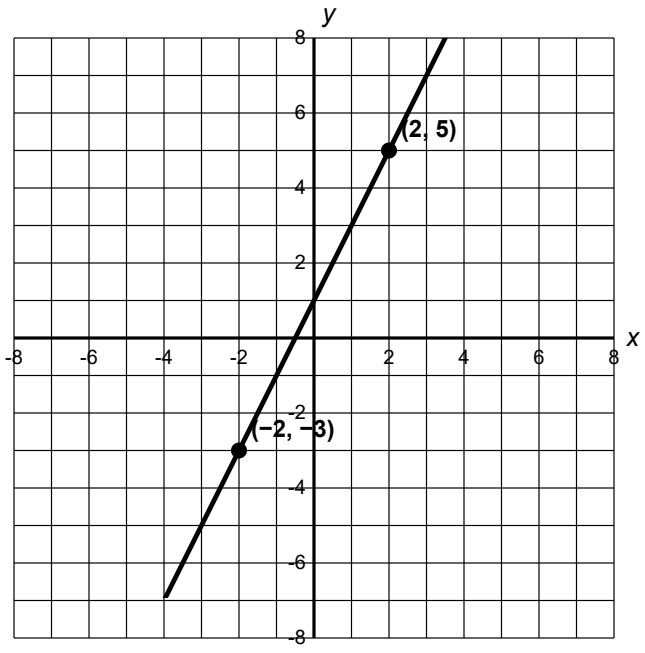
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8. What is the equation in slope-intercept form of the line that passes through the point (5, 0) and is parallel to the line represented by $y = -2.4x + 1.6$?

Equation: _____

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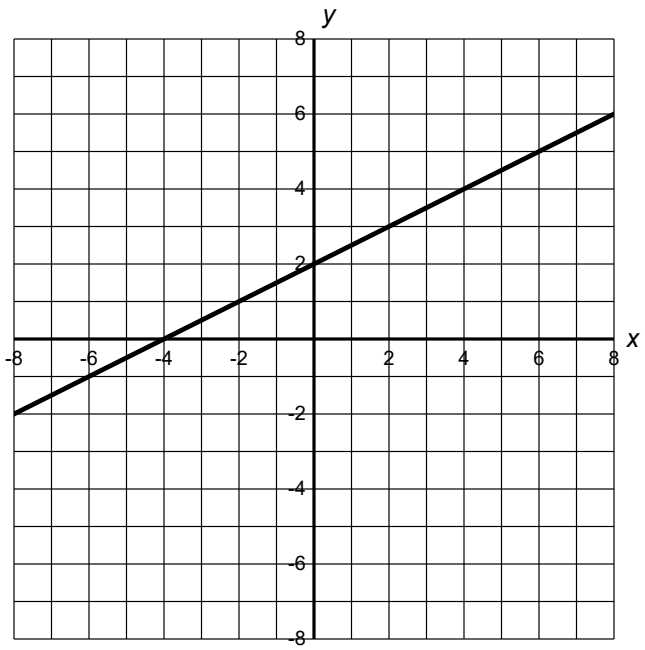
9. Write an equation in point-slope form that represents the graph.



Equation: _____

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10. The graph of a linear function is shown on the coordinate plane.



Which statements describe the graph of the linear function? **Select TWO correct answers.**

F. The graph is increasing.

G. The slope of the line is 2.

H. The *y*-intercept is (0, −4).

J. The zero of the function is −4.

K. The zero of the function is 2.

Part 2 · Extra practice (Questions 11–14)

11. Let $f(x) = x - 1$, $g(x) = f(2x)$, and $h(x) = 2f(x)$. Which statement is true?

A. $g(x)$ and $h(x)$ are the same function.

B. $g(x)$ and $h(x)$ have the same slope but different y-intercepts.

C. $g(x)$ and $h(x)$ have different slopes but the same y-intercept.

D. $g(x)$ and $h(x)$ have different slopes and different y-intercepts.

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12. Which pair of equations represents perpendicular lines?

F. $y = 5x + 1$ and $y = -5x + 2$

G. $y = -\frac{1}{3}x + 4$ and $y = 3x - 2$

H. $y = \frac{4}{5}x$ and $y = \frac{5}{4}x - 1$

J. $y = 2x + 3$ and $y = 2x - 3$

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13. $f(x) = 2x + 1$ and $g(x) = f(x) - 4$. Write the equation of $g(x)$ and describe how the graph of f moves to make the graph of g.

$g(x) =$ _____

Description: _____

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14. A tank holds 300 gallons of water. It drains at 15 gallons per minute until it is empty. Let x be the number of minutes. What is the domain of the function for this situation?

A. $0 \leq x \leq 300$

B. $0 \leq x \leq 20$

C. $0 \leq y \leq 300$

D. $15 \leq x \leq 300$
